

the limitations put on these due to low powered battery cells, their flight time generally limits the usage to a couple of hours. Accordingly, drone operations in India were hit majorly during the lockdown period as no movement of people or machinery was allowed for a couple of months.

“The early stages of Covid-19 crisis were exceedingly difficult for all the businesses. In the drone business, it is all about going to the site, flying drones, and collecting data. The mobilisation restriction nailed down the drone business pretty badly in the first few months,” explains Vipul Singh, founder and CEO, Aarav Unmanned Systems.

He adds that even the existing contracts could not be executed as there was absolute restriction on movements throughout the country. The struck supply chains also made it difficult for upgrading drones as hardware parts were stuck in transit.

“The hangover is still there. However, Covid-19 lockdowns have made people realise the true worth of using drones for different reasons. Drones, in terms of doing remote work, have been able to click on new avenues of growth,” opines Singh.

From reducing manpower in different areas to lessening time taken to do tasks, drones, during the coronavirus lockdowns, were able to showcase their true worth. Singh explains that both government as well as private players have got a taste of what drones are capable of doing. The acceptance of drone technology during the last few months, as per Singh, has shot up. He further notes that while interest in drones is increasing, the amount of business is not increasing in proportion to the same.

Now, while the newspapers were full of breaking stories on how drones were being used to combat coronavirus in different parts of the world, the number of drones being deployed in India for the same was also big. The impact of drones on coronavirus was so big that the United Nations Children’s Fund (UNICEF) published a study explaining how drones can



Agriculture drone spraying water-mixed fertiliser on the sunflower field

be used for combating coronavirus. The study mentions lab sample pick-up and delivery and transportation of medical supplies, aerial spraying of public areas in order to disinfect potentially contaminated places, and public space monitoring and guidance during lockdown and quarantine through drones as three usages reported by nations and media.

“In the face of the global Covid-19 pandemic, there have been reported attempts to utilise drone technology in different scenarios, albeit mainly sporadically, and not at scale,” mentions the UNICEF study.

It continues, “A number of different public safety and law enforcement agencies or organisations (Sierra Leone, Rwanda, China, United States, Spain, Italy, France, UK, India, and others) across the world have deployed drones to surveil the public spaces by gaining a better situational awareness and enforce quarantine by sending messages over a loudspeaker and tracking non-compliant citizens.”

“There has been a lot of involvement of drones in India during the covid times,” says Anurag Joshi, co-founder, INDrone Aero Systems.

He adds, “The outlook of what drones can do and how beneficial they can be were proved to the point during corona time. Despite the low staff government bodies were working with, they were still rolling out permissions for drone flights in India. Another six months and we will see the opportunities around drones going up to good numbers.”

Where do the opportunities lie?

The scope of opportunities associated around drones not only encompass manufacturing, components, and software, but span across training and data collection as well. Flying drones is not an easy task as it requires precision better than flying a manned aircraft. And, as the size of drone increases, the difficulty in flying it increases too.

“The increased demand for drone usage during corona time has increased the demand for drone pilot training as well. We converted all our offices into franchises and that has doubled our revenues as well. A lot of people are taking up drone training as a part of the increased demand as a result of Covid-19,” informs John Livingstone, founder CEO & product architect, Johnnette Technologies Pvt Ltd. Livingstone is also the brains behind Indian Institute of Drones (IID).

While there is no study or research done to assess the number of drone pilots that will be needed in the country during the next five years or so, the number of flights estimated (more than ten million) by different experts hint that India will need at least a million, or may be two million, drone pilots. On the contrary, the number of drone pilots certified by a school (DGCA approved), stood barely at fifteen in November 2020. It is to be noted here that drone flying was legalised in India about two years ago.

Varun Bhatia, Head - Standards, Electronics Skill Sector Council of India (ESSCI), says ESSCI has added new dimensions to the drone pilot training programme in India. The same includes thermal scanning, which can be used to scan 20 to 30 people at a time and record their body temperatures. Drones, as per him, have evolved much more than just flying toys. ESSCI has added battery training, motor system training, flying and aerodynamics, and sensors in its curriculum around drones. ESSCI had